



Paper ID : 260543

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Subject Code: BP701T

Roll No:

BPHARM

(SEM. VII) THEORY EXAMINATION 2025-26

INSTRUMENTAL METHODS OF ANALYSIS – THEORY

TIME: 3 HRS

M.MARKS: 75

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief.

10x 2 = 20

- (a) Define chromophore with an example.
- (b) Recall the name of the radiation source in IR spectroscopy.
- (c) Define the significance of a hollow cathode lamp in AAS.
- (d) What do you mean by isocratic elution chromatography?
- (e) What is the role of the nebulizer in atomic absorption spectroscopy?
- (f) Recall the advantages and disadvantages of gel chromatography
- (g) Define the importance of the guard column in column chromatography.
- (h) Enlist the applications of Thin-layer chromatography.
- (i) What are the main advantages of gas chromatography?
- (j) What is the principle of Ion exchange chromatography?

SECTION B

2. Attempt any *three* of the following:

2 x 10 = 20

- (a) Describe the instrumentation of UV spectroscopy with a labelled diagram.
- (b) Discuss the principle and the factors affecting paper electrophoresis.
- (c) Explain classification, ion exchange resins, properties, and the mechanism of the ion exchange process.

SECTION C

3. Attempt any *one* part of the following:

7 x 5 = 35

- (a) Explain the Theory, Concepts of singlet, doublet, triplet electronic states, internal and external conversions in Fluorescence spectroscopy
- (b) Discuss the Principle, Methodology OF TLC.
- (c) Explain the principle, instrumentation, and applications of Nepheloturbidometry.
- (d) Illustrate the instrumentation of Gel chromatography.
- (e) Explain the instrumentation, derivatization, and temperature programming in gas chromatography.
- (f) Discuss the instrumentation of Affinity chromatography.
- (g) Describe the fundamental modes of vibrations in poly atom and molecules and sample handling techniques in IR spectroscopy